

Nikita Doikov

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Residence: Ithaca, New York, US

RESEARCH INTERESTS

Optimization, Computational Mathematics, Statistics, Machine Learning, Algorithmic Foundations of AI

WORK EXPERIENCE

Assistant Professor at Cornell University , US School of Operations Research and Information Engineering (ORIE)	01.2026 – now
Postdoctoral Researcher at EPFL , Switzerland Machine Learning and Optimization Laboratory , hosted by Martin Jaggi	09.2022 – 12.2025
Postdoctoral Researcher at UCLouvain , Belgium ICTEAM / CORE , hosted by Yurii Nesterov	09.2021 – 08.2022
Intern at Google , Switzerland	Summer 2018
Seminar Teacher at HSE , Russia	Spring 2016, 2017
Intern at Google , Switzerland	Summer 2016
Software Engineer at Yandex , Russia	2015

EDUCATION

PhD in Engineering and Technology , UCLouvain , Belgium Thesis: “ New second-order and tensor methods in Convex Optimization ” Supervisor: Yurii Nesterov	2021
MSc in Computer Science , Higher School of Economics	2017
BSc in Computational Mathematics , Lomonosov Moscow State University	2015

TEACHING AND MENTORING

Instructor at Cornell:

- **ORIE 3300 / 5300**: Optimization I Fall 2026
- **ORIE 6365**: Continuous Optimization: Algorithms and Complexity ([notes](#)) Spring 2026

Leading seminars, practical sessions, and teaching assistance:

- Optimization Models and Methods II at [UCLouvain](#) Spring 2021
- Optimization Methods for Machine Learning at [HSE](#), [CS Department](#) Spring 2017, 2018
- Bayesian Methods in Machine Learning at [HSE](#) 2017, 2018, 2019

Led more than 15 intern, MSc and PhD student research projects

TECHNICAL SKILLS

Advanced: Algorithms and Data Structures, Optimization and Numerical Methods, Machine Learning

Technologies: C++, Python, Matlab, R, SQL, Intel Assembler, UNIX OS architecture, MapReduce, Flume, TensorFlow, PyTorch, JAX

Languages: English (fluent), French (basic), Russian (native)

ACADEMIC SERVICE

Conference reviews: ICML, NeurIPS, COLT, SODA

Journal reviews: Mathematical Programming, SIAM, JOTA, IMA, IEEE

Program Committee: [VOCAL](#) Optimization Conference 2026: Advanced Algorithms

PUBLICATIONS

More than 20 papers published in the leading peer-reviewed journals and conferences on Mathematical Optimization, Statistics, Machine Learning, and AI

Selected Refereed Publications:

<i>Minimizing Quasi-Self-Concordant Functions by Gradient Regularization of Newton Method,</i> N. Doikov , Mathematical Programming Journal	2025
<i>Lower Complexity Bounds for Minimizing Regularized Functions</i> N. Doikov , Optimization Letters Journal	2025
<i>Spectral Preconditioning for Gradient Methods on Graded Non-convex Functions,</i> N. Doikov , S.U. Stich, and M. Jaggi, ICML	2024
<i>Second-order optimization with lazy Hessians,</i> N. Doikov , E.M. Chayti, and M. Jaggi, ICML (Oral Presentation)	2023
<i>Convex optimization based on global lower second-order models,</i> N. Doikov and Yu. Nesterov, NeurIPS (Oral Presentation)	2020
<i>Contracting Proximal Methods for Smooth Convex Optimization,</i> N. Doikov and Yu. Nesterov, SIAM Journal on Optimization	2020

All papers are available on: doikov.com or [Google Scholar](#)

INVITED TALKS, CONFERENCES, AND SCHOOLS

Presented research at 30+ international venues including ICML, NeurIPS, ICLR, AISTATS, SIOPT, ICCOPT

ACHIEVEMENTS AND AWARDS

2025	AISTATS Oral Presentation (2% of all submissions)
2023–2024	Two MSc student theses successfully defended at EPFL under my direct supervision
2023	ICML Oral Presentation (less than 2.4% of all submissions)
2020	ICML Top Reviewer NeurIPS Oral Presentation (less than 1.12% of all submissions)
2019	Best Paper Presentation Award at OBA School NeurIPS Top Reviewers Award
2018	Best Talk Award at Control, Information and Optimization School by B.T. Polyak ICML travel award
2015–2016	Participant of Challenge24 Finals – International 24-hour Programming Contest
2015	Top-200 in Facebook Hacker Cup Moscow festival of sport programming – First degree diploma
2012–2015	University Advanced Scholarship
2014	Moscow festival of sport programming – Second degree diploma All Siberian open programming contest – Second degree diploma Participant of ACM ICPC NEERC – Third degree diploma
2013	Top-600 in Russian Code Cup
2011	Prizewinner of ROI (Russian National Olympiad in Informatics among high school students) – 26th place Gold medal for extraordinary successes in study
2010	Prizewinner of ROI – 90th place